Worksheet 7

due at 23:59 November 2nd (to be discussed November 3rd)

Please include your full name and student ID directly on your homework submission.

1 Action of the QFT on basis states

The action of the QFT on an N-dimensional Hilbert space ($N = 2^n$) with computational basis states $|j\rangle$ is defined through the relation

$$\mathrm{QFT}_N |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i jk/N} |k\rangle .$$

(a) Show that for $n = 1$ ($N = 2$), the matrix QFT_2 is identical to the Hadamard gate H.

(b) Write out the matrix QFT_4 in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Write your matrix in terms of powers of $\omega = e^{2\pi i/4} = i$.

(c) Evaluate $\mathrm{QFT}_8 |000\rangle$ ($j = 0$). Is the resulting state a product state?

2 Circuit representation and complexity

A QFT circuit can be expressed in terms of Hadamard gates and controlled rotation gates R_j, where

$$R_j = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^j} \end{pmatrix}.$$

(a) Give the number of Hadamards and controlled-R_j gates needed to implement a three-qubit QFT circuit. How about four qubits? To keep things simple, disregard the SWAP gates used to invert the bit ordering.

(b) Prove that an n-qubit QFT circuit requires $O(n^2)$ elementary gate operations.

3 Periodicity and Interference

Consider a three-qubit system ($N = 8$) with computational basis states $|j\rangle$ labelled by $j = 0, \ldots, 7$. A three-qubit state is said to have period r if its nonzero amplitudes occur at values separated by r (mod 8). Let's assume the qubits are initialized in the state $|\Psi\rangle$ given by

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|1\rangle + |5\rangle) .$$

(a) What is the period r of this state?

(b) Compute $\mathrm{QFT}_8 |\Psi\rangle$. You may find it easier to evaluate the transform on $|1\rangle$ and $|5\rangle$ separately.

(c) Which computational basis states appear in $\mathrm{QFT}_8 |\Psi\rangle$? How do their indices relate to N and r?

4 Operator Diagonalization

Let X_N be the cyclic shift operator defined by its action on computational basis states $|j\rangle$ according to

$$X_N |j\rangle = |(j+1) \bmod N\rangle , \quad j = 0, 1, \ldots, N-1.$$

Furthermore, let Z_N be the phase operator defined by

$$Z_N = \sum_{k=0}^{N-1} e^{2\pi i k/N} |k\rangle\langle k| .$$

Show that the quantum Fourier transform diagonalizes X_N. In particular, show that

$$\mathrm{QFT}_N X_N \mathrm{QFT}_N^\dagger = Z_N.$$